

PROBLEM SHEET 2

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The following questions are about the language of numbers and strings.

1. Write down the abstract syntax tree for the pre-term `plus(let(len( $x$ );  $i$ . plus( $i$ ;  $n$ )); num[2])`.
2. Assume $\Sigma \stackrel{\text{def}}{=} \{0, 1\}$. Write a program that
 - has a free variable x of type `Str`,
 - appends the string `0110` to x ,
 - computes the length of the compound string, and
 - adds that number to itself.

Your program should not mention the string literal `str[0110]` more than once.

Solution: $x : \text{Str} \vdash \text{let}(\text{len}(\text{cat}(x; \text{str}[0110])); x. \text{plus}(x; x)) : \text{Num}$

3. Produce a typing derivation for the following terms, assuming that $\Sigma \stackrel{\text{def}}{=} \{0, 1\}$.
 - (i) $x : \text{Str} \vdash x : \text{Str}$
 - (ii) $\vdash \text{plus}(\text{num}[1]; \text{num}[1]) : \text{Num}$
 - (iii) $x : \text{Str} \vdash \text{cat}(x; \text{str}[01]) : \text{Str}$
 - (iv) $x : \text{Str}, n : \text{Num} \vdash \text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2]) : \text{Num}$
4. Perform the following substitutions, step-by-step.
 - (i) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[i/x]$
 - (ii) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[\text{num}[0]/n]$
 - (iii) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; n)); \text{num}[2])[i/n]$

Solution: We give the final answer here, not the step-by-step calculation.

- (i) $\text{plus}(\text{let}(\text{len}(i); i. \text{plus}(i; n)); \text{num}[2])$
- (ii) $\text{plus}(\text{let}(\text{len}(x); i. \text{plus}(i; \text{num}[0])); \text{num}[2])$
- (iii) $\text{plus}(\text{let}(\text{len}(x); j. \text{plus}(j; i)); \text{num}[2])$

5. State the cases of the inversion lemma for the following constructs:
 - (i) `len( $e$ )`

(ii) $\text{let}(e_1; x. e_2)$

Solution:

- (i) If $\Gamma \vdash \text{len}(e) : \tau$ then it must be that $\tau = \text{Num}$ and $\Gamma \vdash e : \text{Str}$.
- (ii) If $\Gamma \vdash \text{let}(e_1; x. e_2) : \tau$ there must be τ_1 such that $\Gamma \vdash e_1 : \tau_1$ and $\Gamma, x : \tau_1 \vdash e_2 : \tau$.

6. Prove the weakening lemma for the programming language of numbers and strings.

Solution: The claim is that the rule

$$\frac{\Gamma \vdash e : \tau}{\Gamma, x : \sigma \vdash e : \tau}$$

is admissible. The proof is by induction on the derivation of $\Gamma \vdash e : \tau$.

Case(VAR). Suppose that the derivation is of the form

$$\frac{}{\Gamma, y : \tau \vdash y : \tau} \text{VAR}$$

Then, we see that the following is a valid derivation, which gives us the result.

$$\frac{}{\Gamma, y : \tau, x : \sigma \vdash y : \tau} \text{VAR}$$

Case(NUM). Suppose that the derivation is of the form

$$\frac{n \in \mathbb{N}}{\Gamma \vdash \text{num}[n] : \text{Num}} \text{NUM}$$

Then, we see that the following is a valid derivation, which gives us the result.

$$\frac{n \in \mathbb{N}}{\Gamma, x : \sigma \vdash \text{num}[n] : \text{Num}} \text{NUM}$$

Case(STR). Similar to NUM.

Case(PLUS). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma \vdash e_1 : \text{Num}} \quad \frac{\vdots}{\Gamma \vdash e_2 : \text{Num}}}{\Gamma \vdash \text{plus}(e_1; e_2) : \text{Num}} \text{PLUS}$$

(so in fact $e = \text{plus}(e_1; e_2)$ and $\tau = \text{Num}$). Then by **the inductive hypothesis** we have derivations of $\Gamma, x : \sigma \vdash e_1 : \text{Num}$ and $\Gamma, x : \sigma \vdash e_2 : \text{Num}$. We can combine these into a derivation of the conclusion using the PLUS rule:

$$\frac{\frac{\vdots}{\Gamma, x : \sigma \vdash e_1 : \text{Num}} \quad \frac{\vdots}{\Gamma, x : \sigma \vdash e_2 : \text{Num}}}{\Gamma, x : \sigma \vdash \text{plus}(e_1; e_2) : \text{Num}} \text{PLUS}$$

Case(TIMES). Similar to PLUS.

Case(CAT). Similar to PLUS.

Case(LEN). Similar to PLUS, but with one fewer premise.

Case(LET). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma \vdash e_1 : \sigma_1} \quad \frac{\vdots}{\Gamma, y : \sigma_1 \vdash e_2 : \sigma_2}}{\Gamma \vdash \text{let}(e_1; y. e_2) : \sigma_2} \text{LET}$$

for some σ_1 . (Notice how I have sneakily chosen y to express this rule. By α -conversion the choice of bound variable does not matter.)

By the **inductive hypothesis** we may assume that we have derivations of $\Gamma, x : \sigma \vdash e_1 : \sigma_1$, and $\Gamma, y : \sigma_1, x : \sigma \vdash e_2 : \sigma_2$. Using these derivations we can construct a derivation of the desired judgement by applying the LET rule again:

$$\frac{\frac{\vdots}{\Gamma, x : \sigma \vdash e_1 : \sigma_1} \quad \frac{\vdots}{\Gamma, y : \sigma_1, x : \sigma \vdash e_2 : \sigma_2}}{\Gamma, x : \sigma \vdash \text{let}(e_1; y. e_2) : \sigma_2} \text{LET}$$

7. (*) Complete the proof of substitution from Lecture 4.

Solution: The claim is that the rule

$$\frac{\Gamma \vdash e : \tau \quad \Gamma, x : \tau \vdash u : \sigma}{\Gamma \vdash u[e/x] : \sigma}$$

is admissible. The proof is by induction on the derivation of $\Gamma, x : \tau \vdash u : \sigma$.

(One could try induction on e , but it wouldn't work. Why?)

Case(VAR). Suppose that the typing derivation of u is of the form

$$\frac{}{\Gamma, x : \tau \vdash z : \sigma} \text{VAR}$$

(so in fact u is just a variable z). There are two cases to consider here: either z occurs somewhere in the rest of the context Γ , or the variables z and x are the same.

- If $z \neq x$, then the variable z is somewhere in Γ . By the definition of substitution

$$z[e/x] \stackrel{\text{def}}{=} z$$

Then, the desired derivation is constructed by applying the variable rule:

$$\frac{}{\Gamma \vdash z : \sigma} \text{VAR}$$

- If $z \equiv x$, then we also know that $\sigma = \tau$. By the definition of substitution we have

$$x[e/x] \stackrel{\text{def}}{=} e$$

Then, the desired derivation is just the derivation given by the premise:

$$\frac{\vdots}{\Gamma \vdash e : \tau}$$

Case(NUM). Suppose that the derivation is of the form

$$\frac{n \in \mathbb{N}}{\Gamma, x : \tau \vdash \text{num}[n] : \text{Num}} \text{NUM}$$

(so in fact u is the term $\text{num}[n]$). By the definition of substitution,

$$(\text{num}[n])[x/e] \stackrel{\text{def}}{=} \text{num}[n]$$

Then the following is a valid derivation of the conclusion of the rule.

$$\frac{n \in \mathbb{N}}{\Gamma \vdash \text{num}[n] : \text{Num}} \text{NUM}$$

Case(STR). Similar to NUM.

Case(PLUS). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma, x : \tau \vdash e_1 : \text{Num}} \quad \frac{\vdots}{\Gamma, x : \tau \vdash e_2 : \text{Num}}}{\Gamma, x : \tau \vdash \text{plus}(e_1; e_2) : \text{Num}} \text{PLUS}$$

(so in fact $e = \text{plus}(e_1; e_2)$ and $\sigma = \text{Num}$). By the definition of substitution we have

$$(\text{plus}(e_1; e_2))[e/x] \stackrel{\text{def}}{=} \text{plus}(e_1[e/x]; e_2[e/x]) \quad (1)$$

By the **inductive hypothesis** applied to the two premises, we obtain derivations of the judgments $\Gamma \vdash e_1[e/x] : \text{Num}$ and $\Gamma \vdash e_2[e/x] : \text{Num}$. Using the PLUS rule we can combine these into a derivation

$$\frac{\frac{\vdots}{\Gamma \vdash e_1[e/x] : \text{Num}} \quad \frac{\vdots}{\Gamma \vdash e_2[e/x] : \text{Num}}}{\Gamma \vdash \text{plus}(e_1[e/x]; e_2[e/x]) : \text{Num}} \text{PLUS}$$

But notice that by (1) the subject of this typing derivation is the term $(\text{plus}(e_1; e_2))[e/x]$.

Case(TIMES). Similar to PLUS.

Case(CAT). Similar to PLUS.

Case(LEN). Similar to PLUS, but with one fewer premise.

Case(LET). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma, x : \tau \vdash e_1 : \sigma_1} \quad \frac{\vdots}{\Gamma, x : \tau, y : \sigma_1 \vdash e_2 : \sigma_2}}{\Gamma, x : \tau \vdash \text{let}(e_1; y. e_2) : \sigma_2} \text{LET}$$

for some type σ_1 . By the **induction hypothesis** we get derivations of $\Gamma \vdash e_1[e/x] : \sigma_1$ and $\Gamma, y : \sigma_1 \vdash e_2[e/x] : \sigma_2$. Using the LET rule we obtain a derivation

$$\frac{\frac{\vdots}{\Gamma \vdash e_1[e/x] : \sigma_1} \quad \frac{\vdots}{\Gamma, y : \sigma_1 \vdash e_2[e/x] : \sigma_2}}{\Gamma, x : \tau \vdash \text{let}(e_1[e/x]; y. e_2[e/x]) : \sigma_2} \text{LET}$$

But $(\text{let}(e_1; y. e_2))[e/x] \stackrel{\text{def}}{=} \text{let}(e_1[e/x]; y. e_2[e/x])$ by the definition of substitution.

8. Prove that types are unique, i.e. that for every context Γ and pre-term e there exists at most one τ such that $\Gamma \vdash e : \tau$.

[Hint: assume that there exist two, and prove that they must be the same.]

Solution: The claim is the following: if $\Gamma \vdash e : \sigma$ and $\Gamma \vdash e : \tau$, then $\sigma = \tau$. The proof is by induction on the *first* derivation, namely the derivation of $\Gamma \vdash e : \sigma$.

Case(VAR). Suppose that the derivation is of the form

$$\frac{}{\Delta, x : \sigma \vdash x : \sigma} \text{VAR}$$

(so in fact $\Gamma = \Delta, x : \sigma$, and $e = x$). Then **by the inversion lemma** it must be that $\Delta, x : \sigma \vdash e : \tau$ is also of the same form, i.e. using the VAR rule. But the only way that this can happen is if the type τ matches the type for x , so it must be that $\sigma = \tau$.

Case(NUM). Suppose that the derivation is of the form

$$\frac{n \in \mathbb{N}}{\Gamma \vdash \text{num}[n] : \text{Num}} \text{NUM}$$

(so in fact $e = \text{num}[n]$ and $\sigma = \text{Num}$). Then **by the inversion lemma** the derivation $\Gamma \vdash \text{num}[n] : \tau$ must also have $\tau = \text{Num}$, so $\sigma = \tau$.

Case(PLUS). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma \vdash e_1 : \text{Num}} \quad \frac{\vdots}{\Gamma \vdash e_2 : \text{Num}}}{\Gamma \vdash \text{plus}(e_1; e_2) : \text{Num}} \text{ PLUS}$$

(so in fact $e = \text{plus}(e_1; e_2)$ and $\sigma = \text{Num}$). Then **by the inversion lemma** the derivation $\Gamma \vdash \text{plus}(e_1; e_2) : \tau$ must also have the same shape, and hence $\tau = \text{Num}$, so $\sigma = \tau$.

Case(TIMES). Similar to PLUS.

Case(CAT). Similar to PLUS.

Case(LEN). Similar to PLUS, but with one fewer premise.

Case(LET). Suppose that the derivation is of the form

$$\frac{\frac{\vdots}{\Gamma \vdash e_1 : \sigma_1} \quad \frac{\vdots}{\Gamma, y : \sigma_1 \vdash e_2 : \sigma}}{\Gamma \vdash \text{let}(e_1; y. e_2) : \sigma} \text{ LET}$$

for some σ_1 (so that $e = \text{let}(e_1; y. e_2)$). By the **inversion lemma** we see that the derivation $\Gamma \vdash \text{let}(e_1; y. e_2) : \tau$ must have the form

$$\frac{\frac{\vdots}{\Gamma \vdash e_1 : \sigma'_1} \quad \frac{\vdots}{\Gamma, y : \sigma'_1 \vdash e_2 : \tau}}{\Gamma \vdash \text{let}(e_1; y. e_2) : \tau} \text{ LET}$$

for some type σ'_1 .

We are now in a position where we have obtained a ‘smaller’ sub-derivation $\Gamma \vdash e_1 : \sigma_1$ as well as a derivation $\Gamma \vdash e_1 : \sigma'_1$ for the same pre-term. By the **induction hypothesis** applied to these two judgements we conclude that they have the same type, i.e. $\sigma_1 = \sigma'_1$.

Armed with this fact, we see that we have a ‘smaller’ sub-derivation $\Gamma, y : \sigma_1 \vdash e_2 : \sigma$ as well as a derivation $\Gamma, y : \sigma'_1 \vdash e_2 : \tau$. As $\sigma_1 = \sigma'_1$, we can use the **induction hypothesis** again to conclude that $\sigma = \tau$, which is what we wanted to prove in this case.